

Markscheme

Mathematics: analysis and approaches
Practice question bank

37. (a)

Consider the equation $(\log_3 x)^2 - \log_3(x^4) = 5$.

Find the product of the two values of x satisfying the equation.

$$\text{let } u = \log_3 x: u^2 - 4u - 5 = 0 \quad (M1) \text{ A1}$$

$$(u - 5)(u + 1) = 0 \Rightarrow u = 5 \text{ or } u = -1 \quad (M1)$$

$$x = 3^5 = 243 \text{ or } x = 3^{-1} = \frac{1}{3} \quad \text{A1}$$

$$\text{product} = 243 \times \frac{1}{3} = 81 \quad \text{A1}$$

[5 marks]

(b)

Hence find the value of $\log_9(x_1x_2)$, where x_1 and x_2 are the two solutions from part (a).

$$\log_9(81) \quad (M1)$$

$$= 2, \text{ since } 9^2 = 81 \quad \text{A1}$$

[2 marks]